

Aylar Babaei

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PROFESSIONAL SUMMARY

Bioinformatics Scientist with 2+ years running and troubleshooting genomic pipelines (RNA seq, DNA sequencing, multi cohort harmonization) for translational genomics research. Builds HLA typing and neoantigen prediction workflows and provisions HPC/cloud infrastructure alongside senior bioinformatics staff.

CORE SKILLS & TOOLS

- **Bioinformatics:** RNA seq, DNA sequencing (WGS/WES), multi cohort data harmonization, HLA typing, neoantigen prediction
- **Bioinformatics Tools & Data:** Familiarity with GATK, gnomAD, ClinVar, OMIM, UCSC Genome Browser, NCBI; FASTQ, BAM, CRAM, VCF, MAF, IDAT data formats
- **Scientific & Analytical:** UNIX/Linux, Python, R, Bash/Shell, SQL; biological interpretation of genomic/transcriptomic data; results validation and benchmarking; cross disciplinary collaboration with clinicians and bench scientists
- **Cloud & Workflow Engineering:** CAVATICA/Seven Bridges platform, CWL, Docker, AWS (S3, EC2), HPC (VM provisioning, job scheduling), performance profiling and optimization, Git/GitHub (version control, PR collaboration)
- **Data Operations:** Metadata harmonization, manifest generation, schema validation, QC automation, SOP documentation, data warehousing, basic SQL and relational table structures

EXPERIENCE

Bioinformatics Scientist

Jun 2024 - Present

Children's National Hospital, [BTI Bioinformatics Core](#) · Washington, DC

- Run **multi cohort genomic data harmonization workflows** on raw RNA seq/DNA sequencing data alongside two senior bioinformatics staff; debug and resolve pipeline errors, rerun failed jobs, and document fixes to support reproducibility.
- Provision and manage **HPC/cloud compute environments** (VMs, job scheduling); **profile and optimize pipeline performance** (runtime, memory, resource allocation).
- Built **HLA typing and neoantigen prediction workflows** (RNA2HLA, arcashLA, IEDB based MHC I binding) for immunogenomics analyses.
- Use **AWS cloud compute (S3, EC2) and the CAVATICA/Seven Bridges platform** to support execution of genomic data pipelines.
- **Contribute to production pipeline pull requests**, including methylation based rare tumor subtyping within a national genomics consortium, under senior bioinformatics staff direction, using Git/GitHub for version control.

Graduate Research Assistant

Jan 2023 - May 2024

GWU Computational Biology Institute, [Rahnavard Lab](#) · Washington, DC

- Profiled and quantified LC/MS metabolomics data from iHMP studies to support massSight software development, including quality assessment and biological interpretation of high dimensional datasets.

Graduate Lab Assistant

Mar 2023 - Dec 2023

GWU Cancer Center · Washington, DC

- Supported translational cancer biology studies on CDKN2A/P16 deletion and KRAS activation in mouse intestinal epithelial cells; assisted with Visium spatial transcriptomics generation, tracking, IHC workflows, and genotyping.
- Managed a database of over 2000 laboratory mice, ensuring data accuracy and integrity; maintained breeding, genotype, and health records and assisted with breeding, weaning, and IP injections in support of the mouse model study.

EDUCATION

M.S. Bioinformatics, The George Washington University

Aug 2022 - May 2024

B.S. Chemistry, Shiraz University

Sep 2014 - May 2018